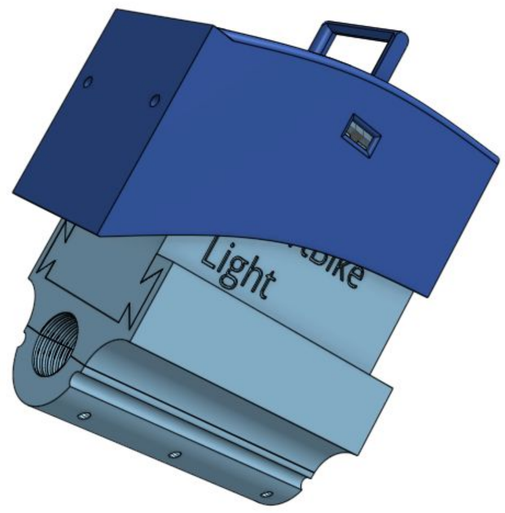
Perception for Smart Bike Lights: Real-Time Embedded Bike Headlight Control Using a Single-Model Convolutional Neural Network



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Introduction

Cyclists experience rapid illumination transitions including direct sunlight, dusk, tunnels, and oncoming vehicles. Manual headlight control increases cognitive load and delays response to glare hazards. While automotive adaptive beam systems exist, they typically rely on heavy object detectors, multi-sensor fusion, and high-power compute, making them unsuitable for low-power bicycle platforms.

This work proposes a vision-only adaptive headlight controller that directly predicts beam state from a single RGB image using a lightweight convolutional neural network.

Contributions

- End-to-end mapping from image → beam state (off/dim/high)
- Policy-derived supervision from BDD100K (no manual beam labeling)
- Embedded hardware integration with ESP32-CAM + GPIO control

Method

Problem Formulation

We model headlight control as a 3-class classification problem:

$$f_\theta(x) \to y \in \{\texttt{off}, \texttt{dim}, \texttt{high}\}$$

Where:

- **off** → daytime (no headlight needed)
- **dim** → night with hazard present
- **high** → night without hazard

Policy-Derived Labeling (BDD100K)

Since BDD100K does not provide beam labels, we derive supervision using:

- Time-of-day attributes
- Hazard bounding boxes
- Central region constraint
- Area threshold (τ = 0.004)

Hazard condition:

$$h(x) = 1 \Leftrightarrow \exists b \in \mathcal{B}(x) : c(b) \in \mathcal{C}, a(b) \geq \tau, \hat{x}(b) \in [\alpha, \beta]$$

Beam label assignment:

- Day → off
- Night + hazard → dim
- Night + no hazard → high

This generates **70K+ training samples** without manual annotation.

System Architecture

Training Configuration

- Input size: 224×224 RGB
- Class-weighted cross entropy with label smoothing
- Oversampling factor K = 3 for minority class
- AdamW optimizer with cosine scheduling
- Mixed precision training
- EMA weight averaging

Model Architecture (Variant E – Final Model)

Lightweight residual CNN:

- Base width: 48 channels
- 4 residual stages
- Group Normalization
- Squeeze-and-Excitation blocks
- DropPath (0.1)

Model size: **4.2M parameters**
Inference time: **20.4 ms (Raspberry Pi 5)**

Temporal Stabilization

To prevent flicker and ensure safety:

- Moving average over N frames
- Hold time H for dim state
- Flicker suppression window F

Dimming is allowed immediately; brightening is delayed.

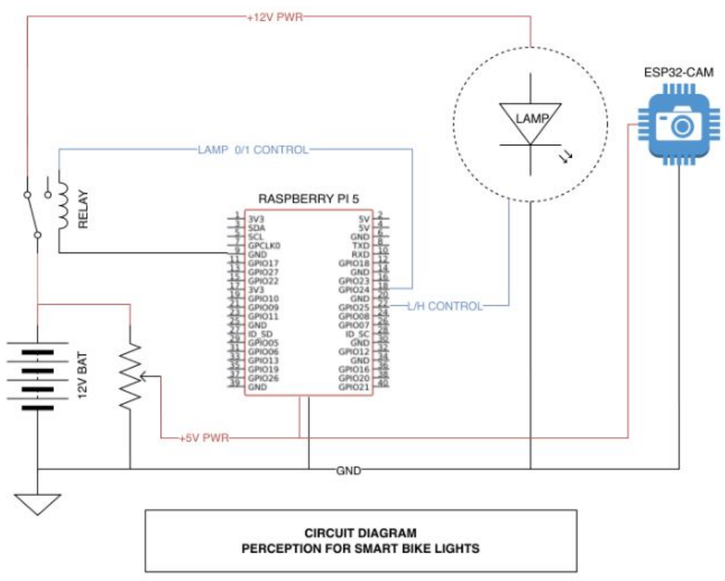
Best deployment settings:

- τ = 0.30
- N = 20
- H = 1.0 s
- F = 2.0 s

Hardware Pipeline

- ESP32-CAM streams MJPEG video
- Raspberry Pi 5 performs inference at 10 Hz
- GPIO controls relay-based headlight switching
- Custom 3D-printed waterproof enclosure

System operates under vibration, pitch changes, and outdoor conditions.



Results

BDD100K Test Performance (18,482 images)

Overall Accuracy: **93.99%**

Class	Precision	Recall	F1
off	99.39%	98.99%	99.19%
dim	94.40%	88.58%	91.39%
high	64.99%	83.03%	72.91%

On-Bike Evaluation (255 Segments)

Safety-critical metric: dimming recall.

Variant E (Single CNN):

- Recall: **80.8%**
- False Positive Rate: 30.6%
- Inference: 20.4 ms

Trade-off observed:

- Modular models → higher recall but unstable thresholds
- Single CNN → improved deployment stability and maintainability

Conclusion

We present a fully embedded adaptive bike headlight system that consolidates perception and control into a single lightweight residual CNN.

Key outcomes:

- Eliminates detector-to-controller threshold tuning
- Scales using policy-derived weak supervision
- Runs at 49 FPS on Raspberry Pi 5
- Demonstrates stable on-bike deployment

The results indicate that **end-to-end policy supervision improves maintainability and robustness more than modular detector pipelines**, particularly under real-world embedded constraints.

Future Work

- Improved twilight transition handling
- Temporal modeling for hazard persistence
- Distance-aware hazard estimation
- Expanded bike-specific dataset collection

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